

Interactive Proofmode for Temporal Logic in Rocq

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Slides:



Context:

We care about liveness and mechanisation properties

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Problem:

Limited support for mechanising temporal properties

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Solution:

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Liveness: something good eventually happens

- ▶ E.g.: Collaborating threads eventually make progress under fair scheduling
- ▶ Often studied in terms of labelled transition system (LTS) traces

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Linear Temporal logic: logic for reasoning about traces

- ▶ $P \vdash Q \triangleq \forall tr. P \text{ } tr \rightarrow Q \text{ } tr$ where $P, Q \in \text{Trace} \rightarrow \text{Prop}$
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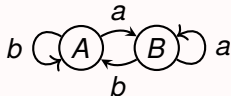
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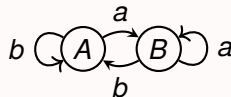
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Most temporal logics are based on modalities, such as

- ▶ Next P : $\circ P \equiv \lambda tr. P(\text{tail } tr)$
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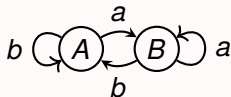
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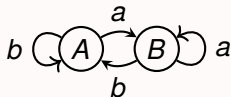
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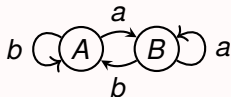
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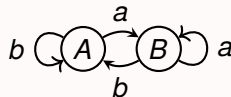
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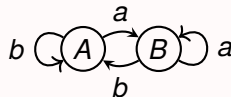
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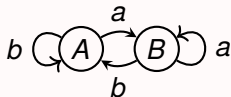
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Mechanisation: machine-checked proofs

- ▶ Trust through rigor; no stone unturned
- ▶ Small trusted computing base
- ▶ Reusable proof infrastructure
- ▶ Increasingly becoming a community expectation
 - ▶ “All results have been mechanised”

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```
From Stdlib Require Import Lia.

Fixpoint fac n :=
  match n with
  | 0 => 1
  | S n' => n * fac n'
  end.

Compute fac 5.

Lemma fac_succ n : fac (S n) = S n * fac n.
Proof. = reflexivity. Qed.

Lemma fac_pos n : fac n > 0.
Proof. =
  induction n. =
  - = simpl. = lia.
  - = rewrite fac_succ. = lia.
Qed.
```

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Definition is_adjoint' {A B : Type} (f : A → B) (g : B → A)
  (issect : g * f == id) (isretr : f * g == id) (a : A) :
  isretr (f a) = ap f (issect' f g issect isretr a). =
Proof. =
  unfold issect'. =
  apply move1_M1. =
  repeat rewrite ap_pp. = rewrite <- concat_p_pp; rewrite <- ap_compose. =
  pose (concat_pA1 (fun b => (isretr b)^) (ap f (issect a))). =
  eapply concat. = 2: { => apply ap2. = reflexivity. = exact e. } =
  cbn. = clear e. =
  pose (concat_pA1 (fun b => (isretr b)^) (isretr (f a))). =
  rewrite <- concat_p_pp. =
  pose (concat_A1p (fun b => (isretr b)) (isretr (f a))). =
  apply move1_Vp in e0. =
  rewrite <- concat_p_pp in e0. = rewrite inv_inv' in e0. =
  rewrite concat_refl in e0. =
  rewrite ap_compose in e0. =
  eapply concat. =
  2: { => apply ap2. = reflexivity. = rewrite concat_p_pp. = eapply
concat. =
  2: { => apply ap2. = eapply concat. =
  2: { => apply ap2. = symmetry. = apply e0. = reflexivity. } =
  symmetry. = apply inv_inv'. = reflexivity. } =
  reflexivity. } =
  repeat rewrite <- ap_compose. =
  cbn. = symmetry. = eapply concat. = refine (ap_pp ((f * g) * f) _)^. =
  rewrite inv_inv. = reflexivity.
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```
Local Lemma try_acquire_spec y lk R :
  {{{ is_lock y lk R }}} try_acquire lk
  {{{ b, RET #b; if b is true then locked y * R else True }}}.
Proof.
  iIntros (Φ) "#Hl HΦ". iDestruct "Hl" as (l ->) "#Hinv".
  wp_rec. wp_bind (CmpXchg _ _ _). iInv N as ([]) "[Hl HR]".
  - wp_cmpxchg_fail "Hinv" : inv M (lock_inv y l R)
  - wp_pures. iApp "Hl" : V b : bool, (if b then locked y * R else True) => Φ #b
  - wp_cmpxchg_succ "Hl" : ▷ l == #true
  iModIntro. iSp "HR" : ▷ True
  rewrite /locked.
Qed.

WP CmpXchg #l #false #true @ T ~ #N ((V, l) => T ~ #N) => ▷ lock_inv y l R * WP Snd

goal 2 (ID 1116) is:
  "Hinv" : inv M (lock_inv y l R)
  "Hl" : V b : bool, (if b then locked y * R else True) => Φ #b
  "HR" : ▷ l == #false
  "HR" : ▷ (taken y * R)
  WP CmpXchg #l #false #true @ T ~ #N ((V, l) => T ~ #N) => ▷ lock_inv y l R * WP Snd
Proof.
  iIntros (Φ) "#Hl HΦ". iDestruct "Hl" as (l ->) "#Hinv".
  wp_apply (try_acquire_spec y lk R). wp_if. iApply "HΦ"; auto with iFrame.
  - iIntros "[Hlkd HR]". wp_if. iApply ("IH" with "[HΦ]"). auto.
  - iIntros "_". wp_if. iApply ("IH" with "[HΦ]"). auto.
Qed.
```



Problem:

Limited support for mechanising temporal properties
(in Rocq)

Option 1: Going nuclear

- ▶ Give a paper proof / (unverified) translation to model checker
- ▶ Increases the trusted computing base
- ▶ Remove “All results have been mechanised” statement

Mechanising Temporal Properties in Rocq

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Option 3: Use existing libraries

- ▶ Browse selection of existing (LTL) libraries in Rocq
- ▶ They focus on metatheory => limited helper lemmas / tactic support

Solution:

Interactive Proofmode for Temporal Logic in Rocq

Introducing Modal Temporal Logic Proofmode (MoTeL)

- ▶ Proofmode context for temporal logic

```
Lemma example (P Q : tProp) :  
  □ (P → ◊ ◊ Q) ⊢ ◊ P → ◊ ◊ Q.
```

```
Proof.
```

```
  iIntros "#HPQ HP".
```

```
  iMod "HP".
```

```
  iDestruct ("HPQ" with "HP") as "HQ".
```

```
  iApply "HQ".
```

```
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- L : Type  
- Rel : S → L → S → Prop  
- P, Q : tProp
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```
-UUU:%%-D F1 *goals*
```

```
No more goals.
```


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- ▶ Primitive Next ($\circ$) modality (even for terminated traces)
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Support for least and greatest fixpoints

- ▶ Logic-level fixpoints with fixpoint theorems for free
- ▶ E.g. $\Diamond P \triangleq \mu X. P \vee \circ X$ $\text{every_other } P \triangleq \nu X. P \wedge \circ \circ P$

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- ▶ Derived Globally ($\Box$) modality via Next

Support for least and greatest fixpoints

- ▶ Logic-level fixpoints with fixpoint theorems for free
- ▶ E.g. $\Diamond P \triangleq \mu X. P \vee \circ X$ $\text{every_other } P \triangleq \nu X. P \wedge \circ \circ P$

Infrastructure for instantiation with custom LTS models

- ▶ Reflect LTS stepping relation as logic-level axioms
- ▶ Extract meta-level theorems about traces

Contributions

Introducing Modal Temporal Logic Proofmode (MoTeL)

- ▶ Proofmode context for temporal logic
- ▶ Tactics for working with temporal modalities

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Proof of eventual delivery for model of Stenning protocol

- ▶ In an unreliable network with a fair delivery assumption

Modal Temporal Logic Proofmode Demo

Developing the Modal Temporal Logic Proofmode

MoSeL: Modal Separation Logic (Proofmode)

- ▶ “Extensible Modal Framework for Interactive Proofs (in Separation Logic)”
- ▶ Foundation for the Iris Proof Mode (facilitating 150+ papers since 2015)
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Key features of MoSeL

- ▶ Managed proof context for spatial and persistent propositions
- ▶ Extensible handling of modality-based reasoning via typeclasses
- ▶ Least and greatest monotone fixpoint constructions
- ▶ (Opt-in) step-indexing and separation logic
- ▶ Coming to Lean!

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Key insights

- ▶ Modality laws of Iris modalities resemble LTL modalities
- ▶ Separation logic and other unused features do not interfere

To Conclude

More case studies

- ▶ Looking for input and collaboration opportunities

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Temporal logic beyond linear

- ▶ Computation tree logic (CTL)
- ▶ μ -calculus $\rightarrow$ derived LTL and CTL

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Temporal separation logic

- ▶ Conceptually stimulating
- ▶ Practically unclear what it even means
- ▶ MoTeL might make it more feasible to test the waters

Support for mechanisation of temporal properties is warranted

- ▶ Currently limited support
- ▶ ITP-level solution preserves small TCB

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MoSeL seems mature enough to handle new domain logics

- ▶ Relatively easy to instantiate
- ▶ Framework can handle functionality beyond original intent
- ▶ I encourage trying it out!

$$\vdash \Box(\textit{Question} \rightarrow \circ \Diamond \textit{Answer})$$

Slides available at:



Rocq code available at:

